INTERNATIONAL SEARCH REPORT

International application No
PCT/EP2022/062922

A. CLASSIFICATION OF SUBJECT MATTER
INV. F21S6/00 F21V7/00 F21V23/04
ADD. F21Y115/10 F21Y103/10

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)
F21S F21Y F21V H01H H03K

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)
EPO-Internal, WPI Data

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
A	DE 10 2017 208999 A1 (VOLKSWAGEN AG [DE]) 29 November 2018 (2018-11-29) paragraph [0027] - paragraph [0066] figures 1-6	1-14
A	US 2017/248289 A1 (VASYLYEV SERGIY [US]) 31 August 2017 (2017-08-31) paragraph [0044] - paragraph [0189] figure 11 paragraph [0093]	1-14
A	WO 2012/062347 A1 (OSRAM AG [DE]; HADRATH STEFAN [DE]; MUSCHAWECK JULIUS [DE]) 18 May 2012 (2012-05-18) page 10, line 19 - page 17, line 13 figures 1,2,3a,3b,4b,	1-14

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[x] Further documents are listed in the continuation of Box C.

[x] See patent family annex.

* Special categories of cited documents :

"A" document defining the general state of the art which is not considered to be of particular relevance

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"O" document referring to an oral disclosure, use, exhibition or other means

"P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance;; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance;; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

"&" document member of the same patent family

Date of the actual completion of the international search	Date of mailing of the international search report
5 September 2022	21/09/2022

Name and mailing address of the ISA/	Authorized officer
European Patent Office, P.B. 5818 Patentlaan 2 NL - 2280 HV Rijswijk Tel. (+31-70) 340-2040, Fax: (+31-70) 340-3016	Blokland, Russell

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International application No
PCT/EP2022/062922

C(Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
A	GB 2 274 160 A (FORD MOTOR CO [US]) 13 July 1994 (1994-07-13) page 4, line 9 - page 9, line 25 figures 1-4	1-14
A	EP 2 982 541 A2 (VOLKSWAGEN AG [DE]) 10 February 2016 (2016-02-10) paragraph [0022] - paragraph [0034] figures 5-8	1-14
A	WO 2009/116778 A2 (HANWOOL INFORMATION TECH CO LT [KR]; BAEK WOON HO [KR] ET AL.) 24 September 2009 (2009-09-24) paragraph [0027] - paragraph [0078] figures 1-15	1-14

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INTERNATIONAL SEARCH REPORT

Information on patent family members

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